



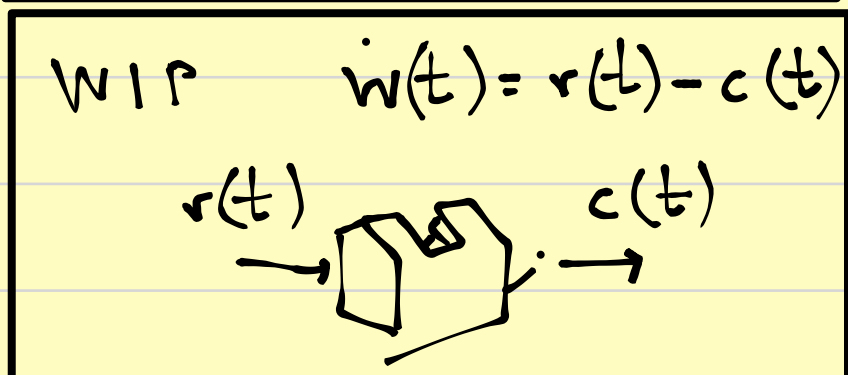
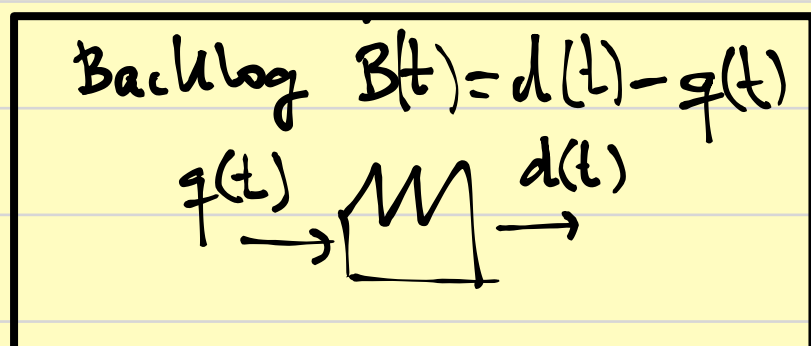
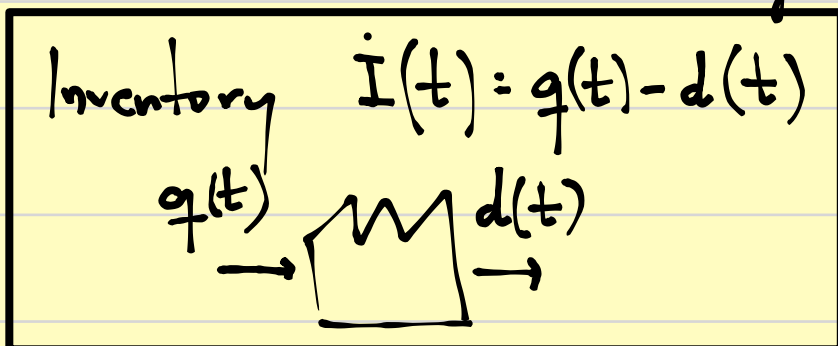
Logistics Replenishment Dynamics with Laplace

BASE STOCK · REORDER POINT · KANBAN · CONWIP · EDQ

Notation:

- $I(t)$ Inventory deviation
- $B(t)$ Backlog
- $W(t)$ Work in Process (WIP)
- $d(t)$ demand or withdrawal rate
- $q(t)$ replenishment rate
- $r(t)$ release rate
- $c(t)$ completion rate
- I^*, W^* target values

The core state balances in logistics:



- Base stock & Reorder Point as feedback loops

Let $I(t)$ be the inventory deviation from the desired level

Use a PROPORTIONAL REPLENISHMENT rule:

$$q(t) = K [I^*(t) - I(t)] \quad (1)$$

Interpret: replenishment rate is proportional (K) to the inventory Δ

- If inventory is below target, then $I^* - I > 0$
so replenishment is increased.

- Combine (1) with inventory balance $\dot{I}(t) = q(t) - d(t)$

- We insert the policy (1):

$$\dot{I}(t) = K [I^*(t) - I(t)] - d(t)$$

- Therefore:

$$\dot{I}(t) + K I(t) = K I^*(t) - d(t)$$

DYNAMIC
REPLENISHMENT

- If there is no demand disturbances $\rightarrow d(t) = 0$, then

$$\dot{I}(t) + K I(t) = K I^*(t)$$

laplace: (with zero initial conditions)

$$s I(s) + K I(s) = K I^*(s)$$

So: $\frac{I(s)}{I^*(s)} = \frac{K}{s + K}$

Transfer function: how many times the actual inventory is contained in the target inventory

- Suppose there is a step target change

$$I^*(t) = \Delta I^* u(t)$$

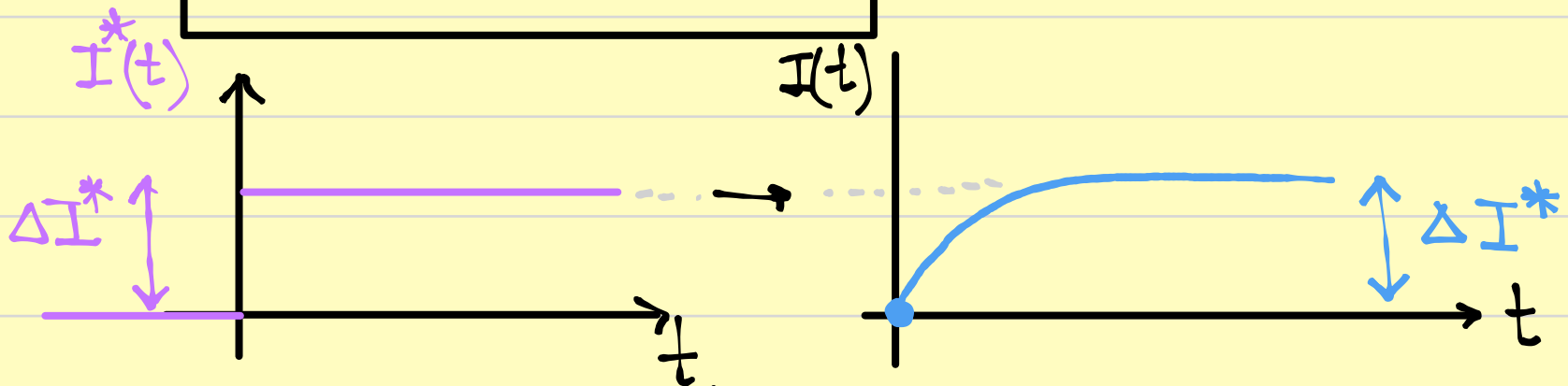
- Then

$$I^*(s) = \frac{\Delta I^*}{s}$$

$$I(s) = \frac{k}{s+k} \cdot \frac{\Delta I^*}{s} = \Delta I^* \cdot \frac{1}{1 + \frac{s}{k}}$$

• Laplace Inverse:

$$I(t) = \Delta I^* (1 - e^{-kt})$$



This is the standard 1st order adjustment of inventory.

• Demand disturbance response with a constant target:

• We have: $I(t) + k I(t) = -d(t)$

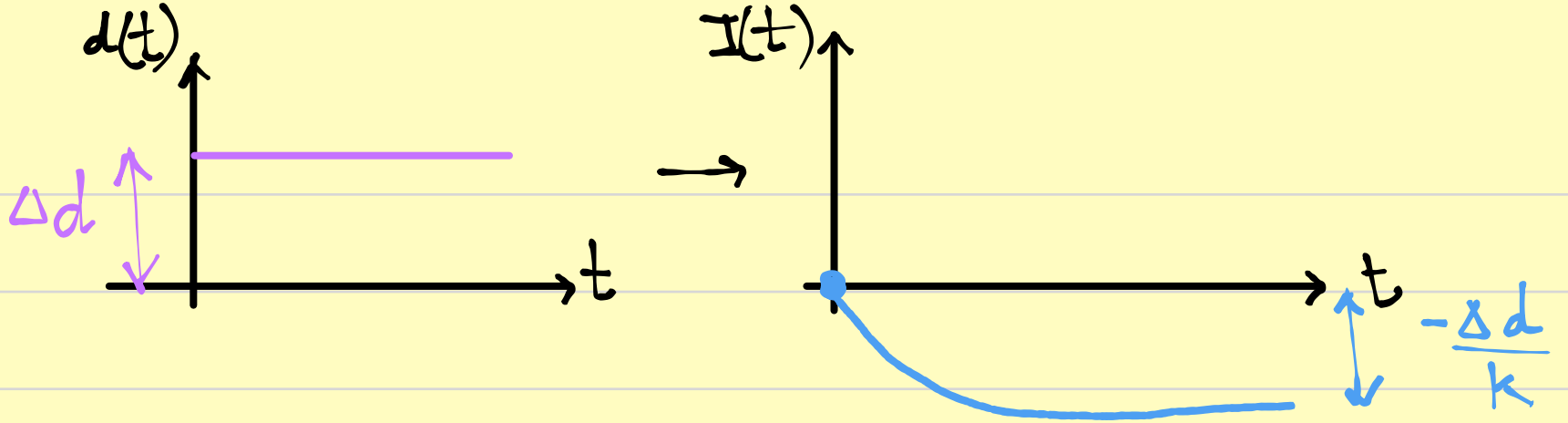
• Laplace: $\frac{I(s)}{D(s)} = \frac{-1}{s+k}$

• Demand step $d(t) = \Delta d \cdot u(t) \rightarrow D(s) = \frac{\Delta d}{s}$

• Therefore: $I(s) = \frac{-\Delta d}{s(s+k)}$

• Hence: $I(t) = -\frac{\Delta d}{k} (1 - e^{-kt})$

Logistics Interpretation: a sudden increase in demand Δd causes an inventory drop



A larger replenishment gain k reduces the long run deviation: if k is big, then $-\frac{\Delta d}{k}$ is small.

• KANBAN

withdrawal $\rightarrow$ replenishment signal $\rightarrow$
 $\rightarrow$ produce/replenish $\rightarrow$ stock refill.

Replenishment does not occur instantly.

There is a lead time.

We model the lag as a first order response of the replenishment rate $q(t)$ to the withdrawal demand $d(t)$.

KANBAN lag model: $T_k \dot{q}(t) + q(t) = d(t)$

.. T_k is "k"

T_k is an effective replenishment TIME constant

Laplace:

$$T_k \cdot s Q(s) + Q(s) = D(s) \rightarrow \frac{Q(s)}{D(s)} = \frac{1}{T_k s + 1}$$

$$\text{Inventory balance: } I(t) = q(t) - d(t) \xrightarrow{d} I(s) = \frac{Q(s) - D(s)}{s}$$

$$I(s) = \frac{\frac{D(s)}{\tau_k s + 1} - D(s)}{s} \rightarrow \frac{I(s)}{D(s)} = \frac{\frac{1}{\tau_k s + 1} - 1}{s}$$

$$\frac{I(s)}{D(s)} = \frac{1 - (\tau_k s + 1)}{s(\tau_k s + 1)} = \frac{-\tau_k s}{s(\tau_k s + 1)} = \frac{-\tau_k}{(\tau_k s + 1)}$$

$$\boxed{\frac{I(s)}{D(s)} = \frac{-\tau_k}{\tau_k s + 1}}$$

• Step increase in the withdrawal:

Suppose $d(t) = \Delta d u(t) \xrightarrow{d} D(s) = \frac{\Delta d}{s}$

$$I(s) = -\frac{\tau_k}{\tau_k s + 1} \cdot \frac{\Delta d}{s}$$

• Inverse Laplace: d^{-1}

$$\boxed{I(t) = -\tau_k \Delta d (1 - e^{-t/\tau_k})}$$

A sudden increase in demand consumes stock during the replenishment delay.
The final inventory loss is $\tau_k \cdot \Delta d$

• CONWIP (constant work in process)

• There is a limit in WIP for the whole process.

• Once this limit has been reached, a withdrawal signal

is triggered.

• The goal here is to keep inventory (WIP) constant.

• State balance: $\dot{w}(t) = r(t) - c(t)$

• Release rule conwip: $r(t) = c(t) + k(w^*(t) - w(t))$

• So: $\dot{w}(t) = c(t) + k(w^*(t) - w(t))$

• Hence: $\dot{w}(t) = k(w^*(t) - w(t))$

• Laplace: $w(s)[s+k] = w^*(s) \cdot k \rightarrow \boxed{\frac{w(s)}{w^*(s)} = \frac{k}{s+k}}$

• Step response in the target: $w^*(t) = \Delta w^* u(t)$

• We get:

$$w(t) = \Delta w^* (1 - e^{-kt})$$

The ideal conwip regulation is a 1st order WIP reg. loop.

• LITTLE'S LAW.

$$\boxed{WIP = TH \cdot CT}$$

$$\begin{array}{ccc} \text{Work in process} & = & \text{Throughput} \cdot \text{Cycle Time} \\ \text{[Parts]} & & \text{[Parts/Time]} \quad \text{[Time]} \end{array}$$

This is true for ANY process.

$$\underbrace{WIP}_{\text{WIP}} \xrightarrow{CT} TH$$

$$CT = \frac{WIP}{TH} \equiv TH = \frac{WIP}{CT}$$



FACTORY PHYSICS (Hopp & Spearman)